

1. Administrative & Study Identification

**Full Study Title:** A Randomized Double-Blind Phase IIA Study Evaluating the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Crovalimab as Adjunct Treatment in Prevention of Vaso-Occlusive Episodes (VOE) in Sickle Cell Disease (SCD)  
**Short Title/Acronym:** CROSSWALK-c **Protocol Number:** B042451 (or 2020-004839-25) **NCT Number:** NCT05075824 **Posting ID:** 557064799952941710 **Sponsor/Company Name:** Hoffmann-La Roche (Roche) **Investigational Product:** Crovalimab **Phase of Development:** Phase IIa (PHASE2) **Trial Status:** Active but not currently recruiting (ACTIVE\_NOT\_RECRUITING) **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

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2. Study Synopsis & Rationale

**2.1 Study Objective** **Primary Objective:** To evaluate the efficacy, safety, and pharmacokinetics of crovalimab compared with placebo as an adjunct treatment in preventing medical facility vaso-occlusive episodes (VOEs) in participants with sickle cell disease (SCD) over a 48-week treatment period. **Secondary Objectives:** To evaluate the annualized rate of acute chest syndrome, the annualized rate of home VOEs, the annualized rate of uncomplicated medical facility VOE, changes in urinary albumin-creatinine ratio, tricuspid regurgitant jet velocity, and Patient-Reported Outcomes Measurement Information System (PROMIS)-Fatigue score. Safety, pharmacokinetics (PK), pharmacodynamics (PD), and immunogenicity are also evaluated.

**2.2 Background & Rationale** Sickle cell disease is caused by a mutation resulting in hemoglobin S, which polymerizes upon deoxygenation, distorting red blood cells into a sickle shape. These abnormal cells cause microvascular occlusions (VOEs) leading to severe pain and target organ damage. Accumulating nonclinical data suggest that complement dysregulation plays a multimodal role in SCD pathophysiology. Crovalimab is a novel anti-complement C5 monoclonal antibody designed to block the complement system, thereby potentially reducing hemolysis and easing the frequency and severity of painful VOE crises.

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3. Study Design & Methodology

**3.1 Design Framework** **Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1

**3.2 Planned Enrollment & Duration Target** **N**: 90 children and adults **Number of Sites**: Global (enrolling across 6 countries) **Estimated Study Start**: 2022 **Estimated Primary Completion**: July 2024

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Age 12 to 55 years; male or female with body weight  $\geq 40$  kg. 2. Confirmed diagnosis of sickle cell anemia: homozygous hemoglobin S (HbSS) or sickle cell  $\beta^0$  thalassemia (HbS $\beta^0$ ). 3. Experienced two or more ( $\geq 2$ ) to  $\leq 10$  documented VOs in the 12 months prior to randomization. 4. If receiving concurrent SCD-directed therapy, the participant must have been on a stable dose for a minimum of 3 months prior to study enrollment, with no plans to modify dosing throughout the study duration (other than for safety reasons). 5. If receiving erythropoietin, the participant must have been prescribed this medication for the preceding 3 months and be dose-stabilized for at least 3 months prior to enrollment. 6. Vaccination against *Neisseria meningitidis* serotypes A, C, W, and Y, and vaccinations against *Haemophilus influenzae* type B and *Streptococcus pneumoniae*. 7. Participants vaccinated (partially or in full) against SARS-CoV-2 with a locally approved vaccine are eligible  $\geq 3$  days after inoculation. 8. Adequate hepatic and renal function. 9. For women of childbearing potential: agreement to remain abstinent or use contraception during the treatment period and for 10.5 months after the final dose of study treatment.

**4.2 Exclusion Criteria** 1. History of hematopoietic stem cell transplant. 2. Participating in a chronic transfusion program and/or planning on undergoing an exchange transfusion during the duration of the study. 3. History of hypersensitivity, allergic, or anaphylactic reactions to any ingredient contained in the study treatment. 4. Received active treatment on another investigational trial within 28 days (or within five half-lives of that agent, whichever is greater) prior to screening visit, or plans to participate in another investigational drug trial. 5. Hemoglobin  $< 6$  g/dL. 6. Known or suspected hereditary complement deficiency. 7. Active systemic bacterial, viral, or fungal infection within 14 days before first drug administration, or presence of fever ( $\geq 38^\circ\text{C}$ ) within 7 days before first drug administration. 8. Immunized with a live attenuated vaccine within 1 month before first drug administration. 9. Pregnant or breastfeeding, or intending to become pregnant during the study or within 10.5 months after the final dose of study treatment. 10. Known HIV infection with documented CD4 count  $< 200$  cells/ $\mu\text{L}$  within 24 weeks prior to screening. 11. History of *N. meningitidis* infection within the prior 6 months.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** Weight-based tiered doses (e.g., 1000 mg for 40–100 kg, 1500 mg for  $\geq 100$  kg). Intravenous (IV) loading dose on Day 1, followed by subcutaneous (SC) doses on Day 2 and weekly during Weeks 2–4. Thereafter, monthly maintenance SC doses for Weeks 5–49. **Route of Administration:** Intravenous (IV) for initial dose, followed by Subcutaneous (SC) injections. **Duration of Treatment:** 48 weeks of active treatment (total study follow-up approximately 91 weeks).

**5.2 Concomitant & Prohibited Therapy Permitted:** Concurrent SCD-directed therapies (e.g., hydroxyurea, L-glutamine, crizanlizumab, voxelotor) and erythropoietin, provided the patient has been on a stable dose for a minimum of 3 months prior to enrollment. **Prohibited:** Immunization with a live attenuated vaccine within 1 month before first drug administration.

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## 6. Study Timeline & Visit Schedule

| Visit ID     | Window (Days) | Key Activities/Procedures                                                           |
|--------------|---------------|-------------------------------------------------------------------------------------|
| Screening    | Day -28 to 0  | Informed Consent, Medical History, Vaccination Checks, Baseline Labs                |
| Baseline     | Day 1         | Randomization, IV Loading Dose (Crovalimab/Placebo)                                 |
| Interim      | Weeks 2 to 49 | SC injections (weekly to W5, then Q4W), VOE Tracking, Efficacy/Safety Labs          |
| End of Study | Week 91       | Final safety follow-up at 24 weeks (clinic) and 46 weeks (telephone) post-treatment |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint** \* Annualized rate of medical facility vaso-occlusive episodes (AVR).

**7.2 Secondary & Exploratory Endpoints** \* Annualized rate of home VOE. \* Annualized rate of uncomplicated medical facility VOE. \* Annualized rate of Acute Chest Syndrome (ACS). \* Change from baseline in urinary albumin-creatinine ratio, tricuspid regurgitant jet velocity, and PROMIS-Fatigue score. \* Safety profiles, PK, PD, and immunogenicity parameters.

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Standard clinical monitoring including infusion-related reactions, injection-site tolerability, and routine safety labs during onsite visits and scheduled follow-up telephone calls. **Serious Adverse Events (SAEs):** Specific monitoring for severe infections (given complement C5 inhibition), with typical 24-hour reporting protocols. **Data Monitoring Committee (DMC):** Independent Data

Monitoring Committee to evaluate ongoing safety data during the trial.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Intent-to-Treat (ITT) Set for the primary efficacy evaluation; Safety Analysis Set for safety parameters and adverse events. **Sample Size Justification:** A target of 90 patients randomized 1:1 provides adequate power to detect a clinically meaningful difference in the annualized VOE rate between the crovalimab and placebo arms. **Handling of Missing Data:** Standard multiple imputation methods and robust generalized linear models (e.g., Negative Binomial regression) for analyzing the annualized count rates of VOEs.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite data verification visits supplemented by remote monitoring. **Audit Trail:** Secure eCRF entry, continuous data clarification workflows, and strict maintenance of the trial's double-blind integrity.

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## 11. Study Identifiers & Governance

**Primary ID:** B042451 **Registry Number:** NCT05075824 **Posting ID:** 557064799952941710 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche) **Study Title:** CROSSWALK-c **Official Regulatory Title:** A Randomized Double-Blind Phase IIA Study Evaluating the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Crovalimab as Adjunct Treatment in Prevention of Vaso-Occlusive Episodes (VOE) in Sickle Cell Disease (SCD) **Trial Status:** ACTIVE\_NOT\_RECRUITING (Active but not currently recruiting)

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## 12. Clinical Framework (Disease/Condition Specifics)

**Primary Condition:** Sickle Cell Disease (SCD) / Sickle Cell **UMLS Preferred Name:** Vaso-Occlusive Crisis **Disease Preferred Name:** Anemia, Sickle Cell **Semantic Type:** Disease or Syndrome **Definition:** A disease characterized by chronic hemolytic anemia, episodic painful crises, and pathologic involvement of many organs. It is the clinical expression of homozygosity for hemoglobin S. **MeSH Code:** D000755 **SNOMED ID:** 49938009 **SNOMED Hierarchy:** 397020000 | Poikilocyte | Poikilocyte (cell) **Diagnosis Threshold:** Confirmed homozygous HbSS or HbS $\beta^0$  with 2-10 VOEs in the previous 12 months **Medical Methodology:** Complement system inhibition (Anti-C5 monoclonal antibody; Crovalimab)

**Optimization Target:** Reduction in the annualized rate of medical facility VOE

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### 13. Study Procedures & Methodology

**Management Pathway:** Double-blind systemic adjunct therapy combined with standard-of-care SCD maintenance. **Education Protocol (HCP):** Training on IV loading and subsequent transition to SC injection administration. **Education Protocol (Patient):** Strict guidelines on recording and reporting home VOE and recognizing signs of severe infections. **Quality Audit Mechanism:** Adherence monitoring for scheduled SC doses and timely VOE documentation.

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### 14. Observation & Follow-Up Schedule

**Total Duration:** Approximately 91 weeks (48 weeks of active treatment + 46 weeks of post-treatment follow-up) **Onsite Visit Cadence:** Day 1, Day 2, Weekly (Weeks 2-5), and Every 4 Weeks (Q4W) until Week 49, with an onsite safety visit 24 weeks after the final dose. **Remote Monitoring Frequency:** Final tele-visit follow-up conducted 46 weeks after the final dose. **Checklist Review Interval:** Regular intervals aligned with Q4W treatment visits.

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### 15. Sign-off & Approval

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|---------------------------------|--------|-----------|---------------|--|------|--|------|--|------|--|------|--|
| Role                            | Name   | Signature | Date          |  | :--- |  | :--- |  | :--- |  | :--- |  |
| <b>Principal Investigator</b>   | [Name] | _____     | [DD-MMM-YYYY] |  |      |  |      |  |      |  |      |  |
| <b>Clinical Operations Lead</b> | [Name] | _____     | [DD-MMM-YYYY] |  |      |  |      |  |      |  |      |  |
| <b>Statistician</b>             | [Name] | _____     | [DD-MMM-YYYY] |  |      |  |      |  |      |  |      |  |